



POLITECNICO DI TORINO

MASTER'S DEGREE IN CYBERSECURITY ENGINEERING

Design and Implementation of an eBPF-based Security Monitoring Tool

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This thesis presents the design and implementation of a security monitoring tool that collects and analyzes network connection data directly from the operating system kernel using eBPF. By leveraging kernel-level instrumentation, the system captures detailed information about network activity with high accuracy and minimal overhead, overcoming the limitations of traditional user-space monitoring approaches.

The proposed tool adopts a modular and efficient architecture to ensure scalability and real-time data collection. The gathered data is structured to support security analysis, enabling the detection of anomalous behaviors such as suspicious outbound connections, unauthorized communication patterns, and potential data exfiltration attempts.

This work demonstrates how kernel-derived network telemetry, combined with eBPF technology, can enhance system visibility and provide a solid foundation for advanced network security monitoring and threat detection.

Acknowledgements

I would like to thank...

Contents

Acronyms	VII
1 Introduction	1
1.1 Background	1
2 State of the Art	2
3 Design	3
4 Implementation	4
5 Evaluation	5
6 Conclusion	6
A Appendix A	7
B Appendix B	8

List of Figures

List of Tables

Acronyms

eBPF	Extended Berkeley Packet Filter
OS	Operating System
NIDS	Network Intrusion Detection System
CVE	Common Vulnerabilities and Exposures
OOB	Out Of Bounds

Chapter 1

Introduction

Insert your introduction text here.

1.1 Background

Background text using eBPF.

Chapter 2

State of the Art

Chapter 3

Design

Chapter 4

Implementation

Chapter 5

Evaluation

Chapter 6

Conclusion

Appendix A

Appendix A

Appendix text here.

Appendix B

Appendix B

Appendix text here.

To my family and friends, for their support.